DAVID MACIAS
Data Scientist

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Versatile data scientist with hands-on experience in data engineering, analytics, and machine learning. In my current role, I've been responsible for the full data workflow—from building pipelines and transforming data to developing models and delivering insights—effectively functioning as a data engineer, analyst, and scientist in one. I'm driven by solving real problems with scalable, data-driven solutions, and I bring a strong background in mathematics, statistics, and numerical analysis.

EXPERIENCE

Data Scientist IDEA Public Schools - Texas 08/2025 - Present

- Created a geographical SSI (Site Suitability Index) score as a solution to help guide school enrollment goals, projections and identify areas for growth.
- Used causal inference to create mapping and effect size tool to guide and advise college graduation trends for matriculated students
- Created testable models to forecast and identify enrollment shortfalls.
- Presented findings to executives with statistical rigor but approachable summarization.

Data Scientist El Paso County, El Paso, TX 04/2024 – 08/2025

- Used LLMs to analyze and summarize hundreds of open-ended question employee satisfaction surveys to generate keyword attributes.
- Completed ETL from various sources, along with data wrangling to make data analysis functional
- Increased data accessibility for non-technical staff

Graduate Researcher New Mexico State University, Las Cruces, NM 08/2023 – 05/2024

- Used LLMs to analyze and summarize hundreds of open-ended question employee satisfaction surveys to generate keyword attributes.
- Completed ETL from various sources, along with data wrangling to make data analysis functional
- Increased data accessibility for non-technical staff

EDUCATION

M.S. Data Analytics New Mexico State University 2022-2024

- Graduated summa cum laude - 3.7+ GPA
- Prominent coursework: **Machine Learning for Algorithm Development**
- Led research team to develop and deploy predictive model for astronomical object classification
- Completed time series analysis to predict drought and precipitation with 95% accuracy.

B.S. Computational Physics University of Texas at El Paso 2016-2020

- Prominent coursework: **Numerical Analysis, Mathematical Methods, Computational Modeling**

DELIVERABLES

Site Selection Causal Inferencing and Predictive Analytics

- Created an end-to-end pipeline that analysis census, demographic, student, state and geographic data to identify areas that correlate with high student enrollment at the K-12 level.
- Used machine learning and causal inference techniques to identify driving factors of enrollment.
- Guided the marketing and recruitment efforts at a local and regional level using linear regression and XGBoost to identify the impact of school closures and transportation on enrollment.

Driving Telematics Analysis and Signal Processing

- Processed raw phone sensor data (magnetometer, accelerometer, gyroscope, Euler angles) to derive usable driver behavior features.
- Modeled and predicted risky driving behaviors using XGBoost and other machine learning methods.
- Developed risk and safety scores for pricing strategies in insurance modeling.

Tech Stack & Skills

Data Storage: Azure, databricks, MySQL, SQL

Data Visualization: Tableau, PowerBI

Data Analysis: Excel, Python, R, SPSS, Apache Spark

Libraries: scikit-learn, tensorflow, sci-py, numpy, pandas, matplotlib, pytorch, SKLearn, XGBoost

Statistical Analysis: Regression, Time-series forecasting, ARIMA, Signal Processing, A/B Testing, Feature Scaling, Machine Learning, Analysis Pipelines, Data Processing, Data Science